

## Mean hitting time of 0 for the diffusion $dX = -\lambda X dt + \sqrt{X(1-X)/N} dW$

Let  $X = (X_t)_{t \geq 0}$  on  $[0, 1]$  solve

$$dX_t = -\lambda X_t dt + \sqrt{X_t(1-X_t)/N} dW_t, \quad X_0 = x \in (0, 1),$$

with  $\lambda > 0$  and  $N > 0$  (the population-size parameter). Let  $\tau_0 := \inf\{t \geq 0 : X_t = 0\}$  and  $u(x) := \mathbb{E}_x[\tau_0]$ .

**ODE.** With  $\sigma^2(x) = x(1-x)/N$  and drift  $b(x) = -\lambda x$ , the generator is  $\mathcal{L} = \frac{1}{2N} x(1-x) \partial_x^2 - \lambda x \partial_x$ , hence

$$\frac{1}{2N} x(1-x) u''(x) - \lambda x u'(x) = -1, \quad u(0) = 0. \quad (1)$$

The right endpoint  $x = 1$  is non-exit (drift inward, diffusion vanishing); for  $2N\lambda < 1$  it is regular and we impose the reflecting condition  $u'(1) = 0$ , for  $2N\lambda \geq 1$  it is entrance and only  $u(1^-) < \infty$  is required.

**Scale and speed.** Since  $2b/\sigma^2 = -2N\lambda/(1-x)$ ,

$$S'(x) = (1-x)^{-2N\lambda}, \quad m(y) = \frac{2}{\sigma^2(y) S'(y)} = \frac{2N(1-y)^{2N\lambda-1}}{y},$$

and

$$S(x) = \int_0^x (1-w)^{-2N\lambda} dw = \begin{cases} \frac{1 - (1-x)^{1-2N\lambda}}{1-2N\lambda}, & 2N\lambda \neq 1, \\ -\ln(1-x), & 2N\lambda = 1. \end{cases}$$

**Solving (1).** With  $g := u'$ ,

$$g'(x) - \frac{2N\lambda}{1-x} g(x) = -\frac{2N}{x(1-x)}.$$

Multiplying by the integrating factor  $(1-x)^{2N\lambda}$  and using that  $g(1^-)$  is finite,

$$(1-x)^{2N\lambda} g(x) = 2N \int_x^1 \frac{(1-y)^{2N\lambda-1}}{y} dy, \quad u'(x) = 2N(1-x)^{-2N\lambda} \int_x^1 \frac{(1-y)^{2N\lambda-1}}{y} dy.$$

Integrating from 0 to  $x$  (using  $u(0) = 0$ ) and applying Fubini,

$$\boxed{\mathbb{E}_x[\tau_0] = \int_0^x S(y) m(y) dy + S(x) \int_x^1 m(y) dy.} \quad (2)$$

**Regularity / numerics.** As  $y \rightarrow 0^+$ :  $S(y) \sim y$ ,  $m(y) \sim 2N/y$ , so  $S(y) m(y) \rightarrow 2N$  (removable). The substitution  $t = (1-y)^{2N\lambda}$  gives the bounded form

$$\int_x^1 m(y) dy = \frac{1}{\lambda} \int_0^{(1-x)^{2N\lambda}} \frac{dt}{1-t^{1/(2N\lambda)}}, \quad (3)$$

with integrand 1 at  $t = 0$  and  $1/x$  at the upper limit. Hence adaptive Simpson on  $[0, (1-x)^{2N\lambda}]$  converges fast.

**Time rescaling.** Equivalently,  $X_t = Y_{t/N}$  where  $dY = -(N\lambda)Y dt + \sqrt{Y(1-Y)} dW$ ; thus  $u(x) = N \cdot u_{\text{base}}(x; \tilde{\lambda} = N\lambda)$ , with  $u_{\text{base}}$  the hitting time for the unit- $N$  SDE.

**Initial condition used in the visualization.** For the Muller's ratchet figures we take

$$X_0 = e^{-\theta} = N^{-\delta}, \quad \theta = \delta \ln N,$$

so  $u(e^{-\theta})$  depends on  $(\psi, \delta, N)$  via  $\lambda = \psi \theta N^{-\beta}$ ,  $\beta = 1 - \delta$ . We evaluate (2) numerically per parameter set using (3) and adaptive Simpson quadrature.